Stepper Library

The servo motor is an easy motor to hook up and does what it does with ease, but what if we want that same kind of positional accuracy in a motor that can spin the full 360? For that, we need to use a stepper motor, which is a type of motor that rather than spinning continuously, rotates in a specific number of degrees or steps. The amount of rotation per step is dependent on the individual stepper; ours has a rotation of 1.8º per step while others can range from as little as 0.9º per step to as much as 30º per step or more.

To control a stepper motor, the internal coils of the motor need to be individually energized in a particular sequence, dependent on the type of stepper motor, either unipolar or bipolar. Unipolar steppers often have five or six wires with a common ground connection and the current flows through each coil in only one direction. Bipolar steppers, on the other hand, only have four wires with no common connection and the current flows through each coil in both directions.

While there are a multitude of drivers available, we are going to use one of two simple chips that only cost a couple dollars each and require no external components to get going. To control a unipolar stepper motor, we will use the ULN2003 Darlington transistor array. A Darlington array is a series of transistor pairs, where a small transistor drives a second transistor to increase its capacity, all on a single integrated circuit or IC. Four individual transistors would have the same effect, although not as convenient. The ULN2003 actually has seven transistor pairs on the chip, although we will only use four of them. By turning each transistor on in order, we can power each coil of the motor causing the motor to spin.

For bipolar stepper motors, we will use the SN754410 dual H-bridge. An H-bridge is a special arrangement of transistors on a chip that is capable of reversing the direction of the current between two pins. When used on standard DC motors, this would change the direction that the motor spins. In the case of a bipolar stepper, alternating the direction of current flow on each coil will create a step causing the motor to turn in a direction determined by the sequence.

Because every stepper is a little different, we will provide two schematics and illustrations for the two types of motor: Figures 9-8 and 9-9 for unipolar and 9-10 and 9-11 for bipolar. Pick the circuit that matches your motor type, being careful to wire the stepper motor coils as shown in the schematic. We have also used a motor that can safely operate from the Arduino board's +5v output. If you are using a motor with a different voltage or higher current requirement, then you should connect an external power supply to the +V Motor pins instead of +5v. For more information on determining the electrical characteristics of stepper motors, you might want to check out the following links:

- `www.makingthings.com/documentation/how-to/stepper-motor`
- `www.cs.uiowa.edu/~jones/step/types.html`
- `www.jasonbabcock.com/computing/breadboard/unipolar/index.html`
- `www.jasonbabcock.com/computing/breadboard/bipolar/index.html`